

Analysis of Variance

It is a technique that will enable us to test for the significance of the difference among more than two sample means when the sample observations are independent, they are randomly selected from the population and the population is normal.

Working Procedure :

1) Null Hypothesis : there is no significant difference in means.
 H_1 : there is significant difference in means.

2) Find N = total no. of observations

T = total value of observations

3) correction factor, $C.F. = \frac{T^2}{N}$

4) Total sum of squares $TSS = (\sum X_1^2 + \sum X_2^2 + \dots) - C.F.$

5) Column sum of squares $SSC = \left(\frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_2} + \dots \right) - C.F.$

where N_i = total no. of observations in each column.

6) Prepare the ANOVA table to calculate P-ratio $SSE = TSS - SSC$

Source of variation	Sum of squares	degree of freedom	mean square	F-ratio
Between columns	SSC	$c-1$	$MSC = \frac{SSC}{c-1}$	$F_c = \frac{MSC}{MSE}$ if $MSC > MSE$
Error	SSE	$N-c$	$MSE = \frac{SSE}{N-c}$	$F_c = \frac{MSE}{MSC}$ if $MSE > MSC$

Total

7) Find table value

8) Cal. value $<$ Table value, accept H_0

Cal. value $\geq$ Table value, reject H_0

Q: A completely randomized design experiment with 10 plots and 3 treatments gave the following results:

Plot No :	1	2	3	4	5	6	7	8	9	10
Treatment:	A	B	C	A	C	C	A	B	A	B
Yield :	5	4	3	7	5	1	3	4	1	7

Analyse the results for treatment effects.

A	B	C	H_0 - no significant diff.
5	4	3	H_1 -
7	4	5	
3	7	1	
1			

X_1	X_2	X_3	Total	X_1^2	X_2^2	X_3^2
5	4	3	12	25	16	9
7	4	5	16	49	16	25
3	7	1	11	9	49	1
1			1	1		
16	15	9	40	84	81	35

$$C = 3, N = 10, T = 40$$

$$CF = \frac{T^2}{N} = 160$$

$$TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 - CF = 84 + 81 + 35 - 160 = 40$$

$$SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_2} + \frac{(\sum X_3)^2}{N_3} - CF = \frac{(16)^2}{4} + \frac{(15)^2}{3} + 3 - 160 = 6$$

N_i = no. of elements in each column

$$SSE = TSS - SSC = 34$$

ANOVA Table:

Source of variation	Sum of Squares	Degree of freedom	Mean Square	F-ratio
Between columns	$SSC = 6$	$C-1 = 3-1 = 2$	$MSC = \frac{SSC}{C-1}$ $= 6/2 = 3$	$MSE > MSC$
Error	$SSE = 34$	$N-C = 10-3 = 7$	$MSE = \frac{SSE}{N-C}$ $= \frac{34}{7}$ $= 4.86$	$F_c = \frac{MSE}{MSC}$ $= \frac{4.86}{3}$ $= 1.62$

$$F_c = 1.62$$

Tab $F_{c,tab}(7, 2) = 19.35$ — F-table

$$F_c < Tab F_c$$

∴ H_0 accepted.